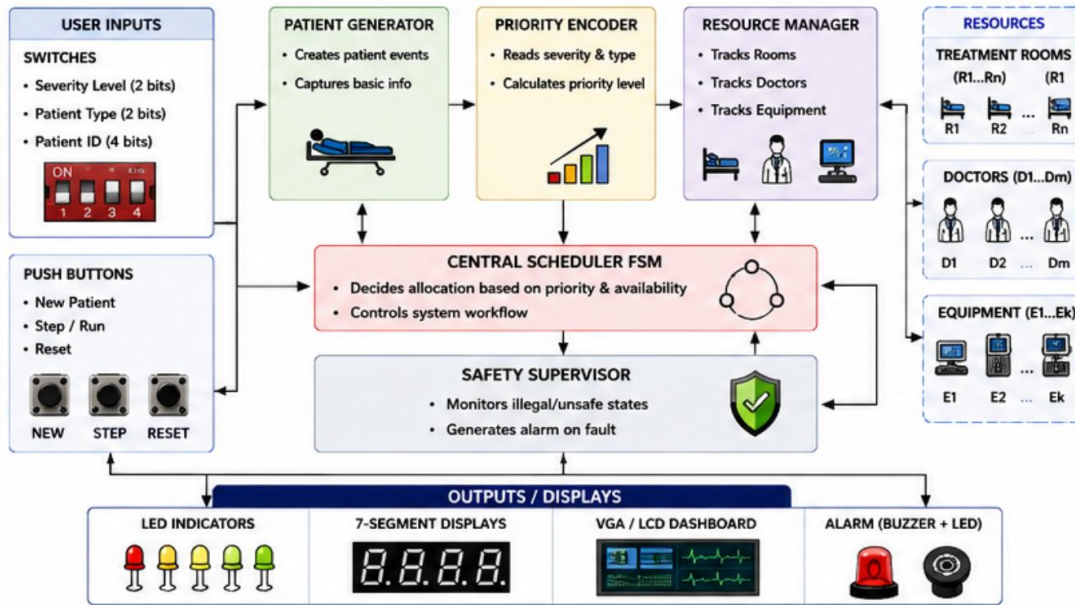


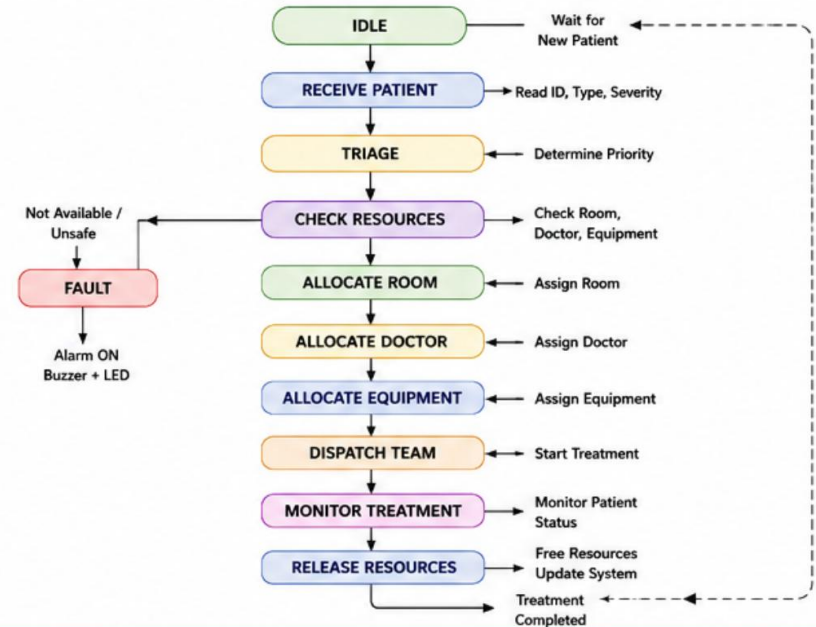
FPGA-BASED SMART HOSPITAL EMERGENCY RESPONSE DISPATCHER

Real-Time Resource Allocation, Safety Monitoring and Visualization Using NEXYS A7 – Artix-7 FPGA

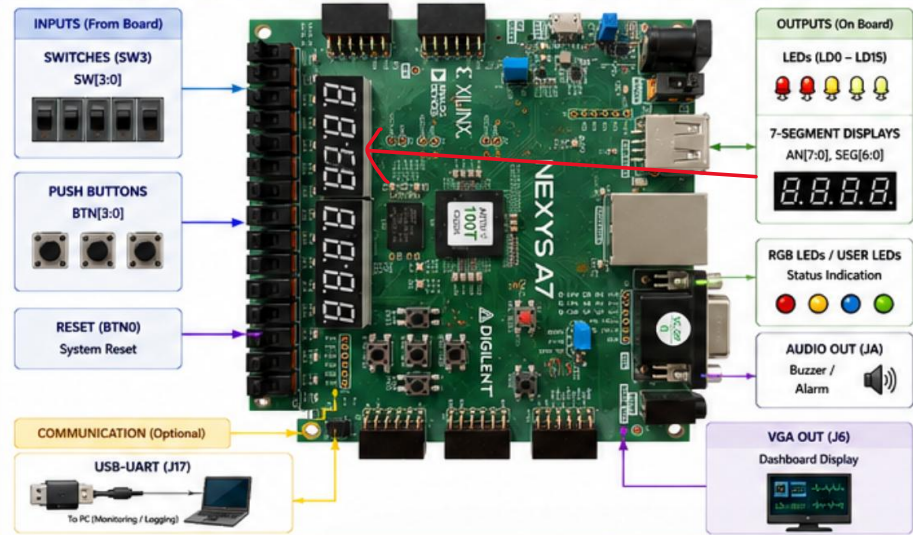
1. SYSTEM ARCHITECTURE



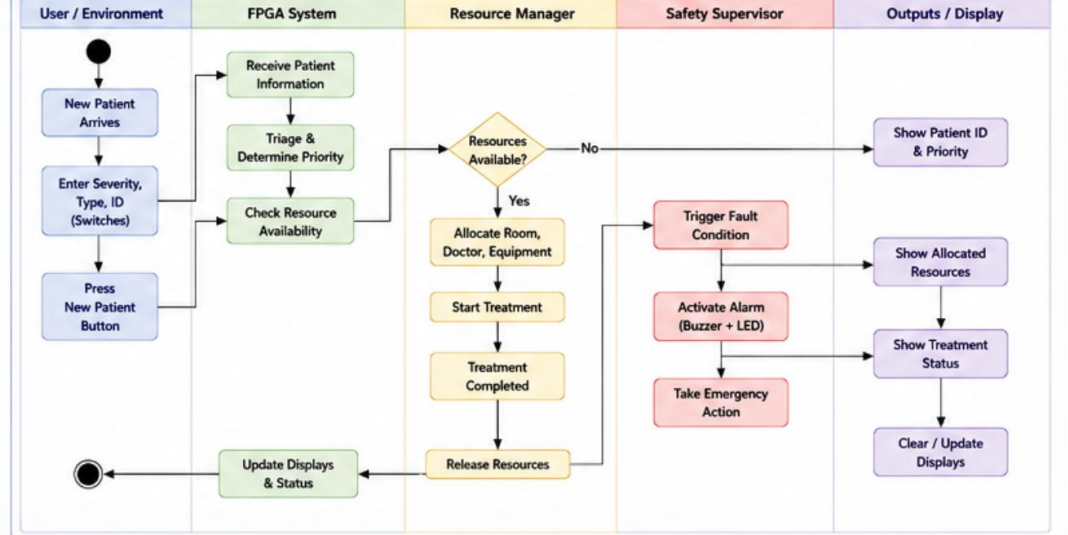
2. SCHEDULER FSM (STATE DIAGRAM)



3. NEXYS A7 – ARTIX-7 POWER BOARD DEMONSTRATION MAPPING



4. ACTIVITY DIAGRAM (SYSTEM WORKFLOW)



FPGA-Based Smart Hospital Emergency Response Dispatcher Using the Digilent Nexys A7 (Artix-7 XC7A100T)

1. Project Overview

The FPGA-Based Smart Hospital Emergency Response Dispatcher is a real-time embedded system implemented entirely in VHDL on the Digilent Nexys A7 FPGA board. The purpose of the system is to simulate the operation of a hospital emergency department where patients arrive with different levels of urgency and must be assigned available resources such as treatment rooms, doctors, and medical equipment.

The FPGA acts as an intelligent hospital dispatcher that receives patient information through board inputs, prioritizes patients according to severity, allocates resources, monitors system safety, and displays all system activities on LEDs, seven-segment displays, UART communication, and optionally a VGA dashboard.

This project combines concepts from:

- FPGA Design
- VHDL
- Embedded Systems
- Cyber-Physical Systems
- Functional Safety
- Real-Time Systems
- Resource Scheduling
- Finite State Machines

The project is intended to demonstrate a realistic safety-critical control system similar to those used in hospitals, airports, industrial automation, and transportation systems.

2. System Objective

The objective of the project is to develop a real-time hospital emergency management system capable of:

- Receiving new patient requests
- Determining patient priority
- Allocating rooms
- Allocating doctors
- Allocating equipment
- Monitoring treatment progress
- Releasing resources after treatment
- Detecting unsafe conditions
- Activating alarms during faults
- Displaying system status on the FPGA board

3. Overall System Architecture

The system consists of the following major modules:

Hospital Dispatcher

|

├— Input Controller

├— Patient Generator

├— Priority Encoder

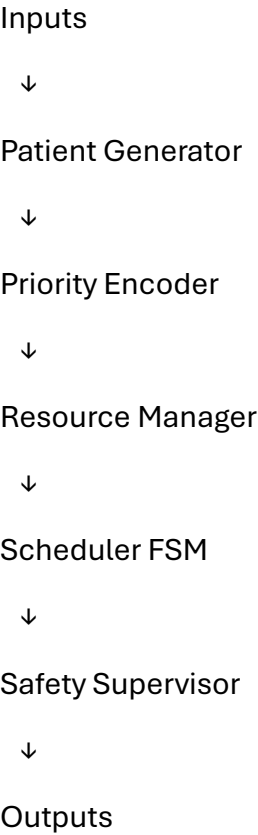
├— Resource Manager

├— Scheduler FSM

├— Safety Supervisor

- └─ LED Controller
- └─ Seven Segment Driver
- └─ UART Interface
- └─ VGA Dashboard (Optional)

The information flow is:



4. User Inputs

The user interacts with the system through the Nexys A7 board.

Switch Inputs

The slide switches represent patient information.

SW0-SW1

Patient Severity

00 = Normal

01 = Urgent

10 = High Priority

11 = Critical Emergency

SW2-SW3

Patient Type

00 = General

01 = Accident

10 = Stroke

11 = Heart Attack

SW4-SW7

Patient Identification Number

0000 – 1111

Patient IDs 0 – 15

Push Buttons

BTN0

System Reset

Returns the entire system to the IDLE state.

BTN1

New Patient

Generates a new patient arrival event.

BTN2

Step FSM

Executes one state transition at a time for demonstration purposes.

BTN3

Emergency Override

Immediately prioritizes critical emergency patients.

5. Patient Generator Module

The Patient Generator creates patient records from the board inputs.

Generated information includes:

Patient ID

Patient Type

Patient Severity

Arrival Time

Example:

Patient ID = 12

Patient Type = Heart Attack

Severity = Critical

The generated information is forwarded to the Priority Encoder.

6. Priority Encoder Module

The Priority Encoder performs triage.

Triage is the process used in hospitals to determine which patient should be treated first.

Example:

Patient A = Critical

Patient B = Normal

Patient C = Urgent

Output order:

Patient A

Patient C

Patient B

The Priority Encoder always gives preference to the highest-severity patient.

7. Resource Manager Module

The Resource Manager keeps track of available hospital resources.

Treatment Rooms

R1

R2

R3

R4

Doctors

D1

D2

D3

D4

Equipment

E1

E2

E3

E4

The module stores occupancy information and continuously updates resource availability.

8. Scheduler FSM

The Scheduler Finite State Machine is the central controller of the system.

The FSM executes the following sequence:

IDLE

↓

RECEIVE_PATIENT

↓

TRIAGE

↓

CHECK_RESOURCES

↓

ALLOCATE_ROOM

↓

ALLOCATE_DOCTOR

↓

ALLOCATE_EQUIPMENT

↓

DISPATCH_TEAM

↓

MONITOR_TREATMENT

↓

RELEASE_RESOURCES

↓

IDLE

If a fault occurs:

CHECK_RESOURCES

↓

FAULT

State Descriptions

IDLE

Waiting for a new patient.

RECEIVE_PATIENT

Reads patient information from switches.

TRIAGE

Calculates patient priority.

CHECK_RESOURCES

Verifies room, doctor, and equipment availability.

ALLOCATE_ROOM

Assigns treatment room.

ALLOCATE_DOCTOR

Assigns doctor.

ALLOCATE_EQUIPMENT

Assigns medical equipment.

DISPATCH_TEAM

Begins treatment.

MONITOR_TREATMENT

Tracks treatment progress.

RELEASE_RESOURCES

Returns all resources to the available pool.

FAULT

Activates alarms and safety procedures.

9. Safety Supervisor

The Safety Supervisor continuously monitors system operation.

The module checks for:

Double Allocation

Two patients assigned to the same room.

Doctor Conflict

One doctor assigned to multiple patients.

Equipment Conflict

One equipment unit assigned multiple times.

Resource Exhaustion

No room available.

No doctor available.

No equipment available.

If any fault is detected:

LED14 = ON

LED15 = ON

Alarm = ON

10. Nexys A7 FPGA Board Mapping

The system is mapped directly to the Digilent Nexys A7 FPGA board.

Switches

SW0-SW1 → Severity

SW2-SW3 → Patient Type

SW4-SW7 → Patient ID

Push Buttons

BTN0 → Reset

BTN1 → New Patient

BTN2 → Step FSM

BTN3 → Emergency Override

LEDs

Room Status

LED0 → Room 1

LED1 → Room 2

LED2 → Room 3

LED3 → Room 4

Doctor Status

LED4 → Doctor 1

LED5 → Doctor 2

LED6 → Doctor 3

LED7 → Doctor 4

Equipment Status

LED8 → Equipment 1

LED9 → Equipment 2

LED10 → Equipment 3

LED11 → Equipment 4

System Status

LED12 → Critical Patient

LED13 → Treatment Active

LED14 → Fault

LED15 → Alarm

11. Seven Segment Display Mapping

The Nexys A7 contains eight seven-segment displays.

AN0 → Patient ID

AN1 → Priority

AN2 → Room Number

AN3 → Doctor Number

AN4 → Equipment Number

AN5 → FSM State

AN6 → Fault Code

AN7 → System Counter

12. UART Communication

The UART interface provides monitoring and debugging.

Example terminal output:

Patient 12 Arrived

Severity = Critical

Room Assigned = R2

Doctor Assigned = D4

Equipment Assigned = E1

Treatment Started

Treatment Completed

Resources Released

13. VGA Dashboard (Optional)

The VGA interface displays a real-time hospital dashboard.

Example:

SMART HOSPITAL DASHBOARD

Patient ID : 12

Priority : Critical

Room : R2

Doctor : D4

Equipment : E1

System State : Treatment

This significantly improves the demonstration quality.

14. Activity Flow

The system operates according to the following workflow:

Patient Arrives

↓

Read Inputs

↓

Generate Patient

↓

Perform Triage

↓

Check Resources



Resources Available?

/ \

YES NO

| |

Allocate FAULT

Resources |

| |

Treatment Alarm

|

Monitor

Treatment

|

Release

Resources

|

Update

Displays

|

Return to IDLE

15. Development Process

Phase 1

Requirements Analysis

Define:

- Patients
 - Rooms
 - Doctors
 - Equipment
 - Safety Requirements
-

Phase 2

System Architecture

Create block diagrams and system decomposition.

Phase 3

FSM Design

Design all states and transitions.

Phase 4

VHDL Module Development

Create:

patient_generator.vhd

priority_encoder.vhd

resource_manager.vhd

scheduler_fsm.vhd

safety_supervisor.vhd

seven_segment_driver.vhd

led_controller.vhd

uart_controller.vhd

Phase 5

Simulation

Create testbenches.

Test:

- Normal patient
- Critical patient
- Resource exhaustion
- Safety violations

Phase 6

Hardware Integration

Create:

top_level.vhd

Connect all modules.

Phase 7

Synthesis and Bitstream Generation

Use Xilinx Vivado to:

Synthesize

Implement

Generate Bitstream

Phase 8

FPGA Demonstration

Program the Nexys A7 board and demonstrate:

Scenario 1

Normal Patient

Scenario 2

Critical Emergency

Scenario 3

Resource Exhaustion

Scenario 4

Emergency Override

Scenario 5

Safety Fault Detection

16. Expected Learning Outcomes

After completing this project, the team will have demonstrated:

- Finite State Machine Design
- VHDL Development
- FPGA Hardware Design
- Real-Time Scheduling
- Resource Allocation Algorithms
- Functional Safety Concepts
- UART Communication
- VGA Display Control
- Verification and Testing
- Cyber-Physical Systems Engineering
- Hardware/Software Co-Design Principles

This makes the FPGA-Based Smart Hospital Emergency Response Dispatcher a highly creative, industrially relevant, and technically challenging FPGA project that can be fully demonstrated on the Digilent Nexys A7 Artix-7 FPGA board.